EXPERIMENT 2

A.SHELL SCRIPT – REVERSE A NUMBER USING while LOOP

AIM: To reverse the digits of an integer using a while loop in a shell script.

SOFTWARE REQUIRED: Oracle VirtualBox, Fedora Linux, Terminal, Bash shell.

PROCEDURE:

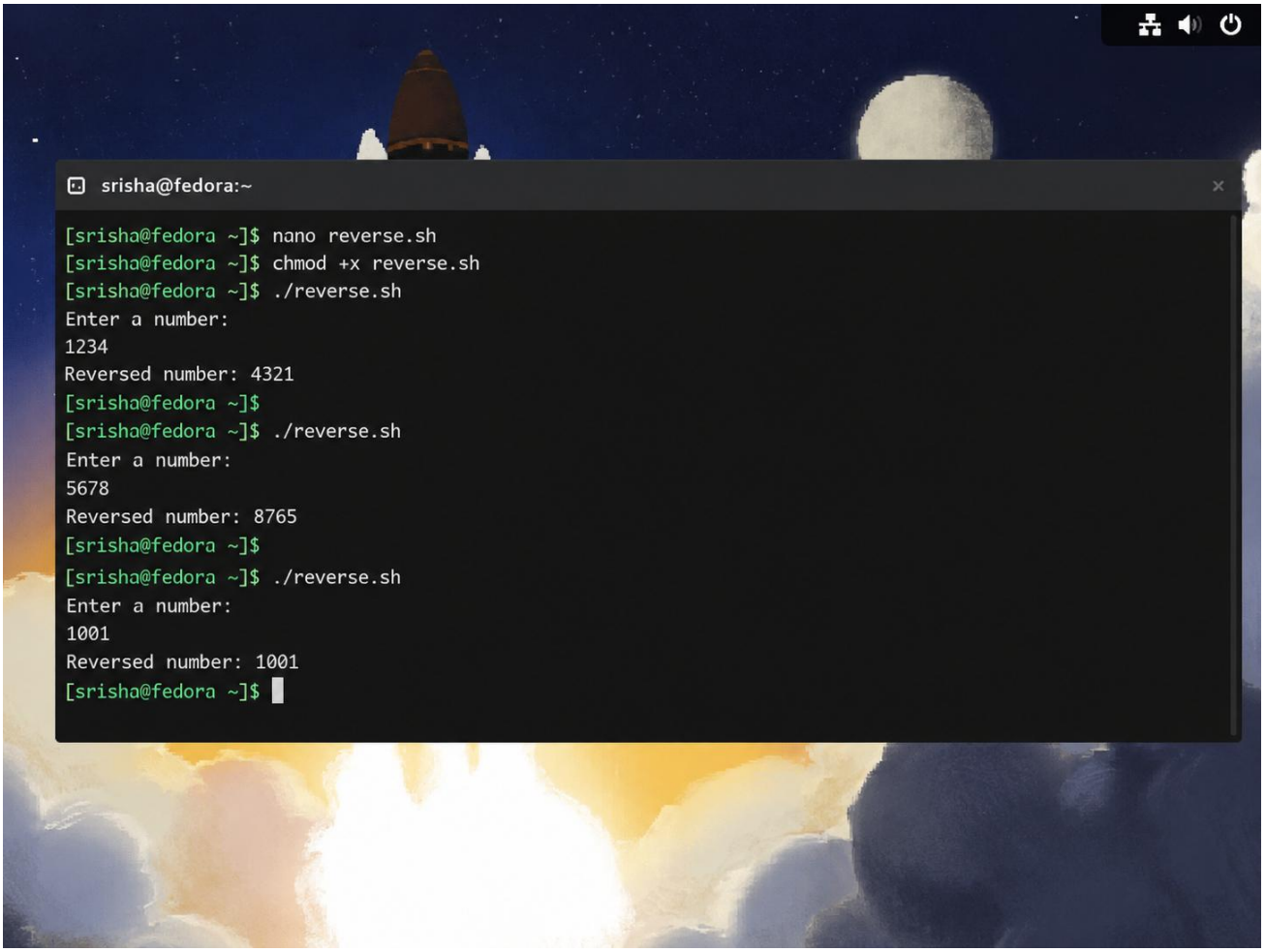
1. Create reverse.sh in the Fedora terminal.
2. Enter and save the script.
3. Run chmod +x reverse.sh followed by ./reverse.sh.
4. Enter a non-negative integer and observe the reversed number.

PROGRAM:

```
echo "Enter a number:"
read num
temp=$num
rev=0
while [ "$temp" -gt 0 ]; do
    digit=$(expr "$temp" % 10)
    rev=$(expr "$rev" \* 10 + "$digit")
    temp=$(expr "$temp" / 10)
done
echo "Reversed number: $rev"
```

OUTPUT:

```
Enter a number:
12345
Reversed number: 54321
```



RESULT: Thus, the script correctly checked whether the entered year was a leap year.

EXPERIMENT 2

B.SHELL SCRIPT – FIBONACCI SERIES USING for LOOP

AIM: To generate the Fibonacci series for a specified number of terms using a for loop.

SOFTWARE REQUIRED: Oracle VirtualBox, Fedora Linux, Terminal, Bash shell.

PROCEDURE:

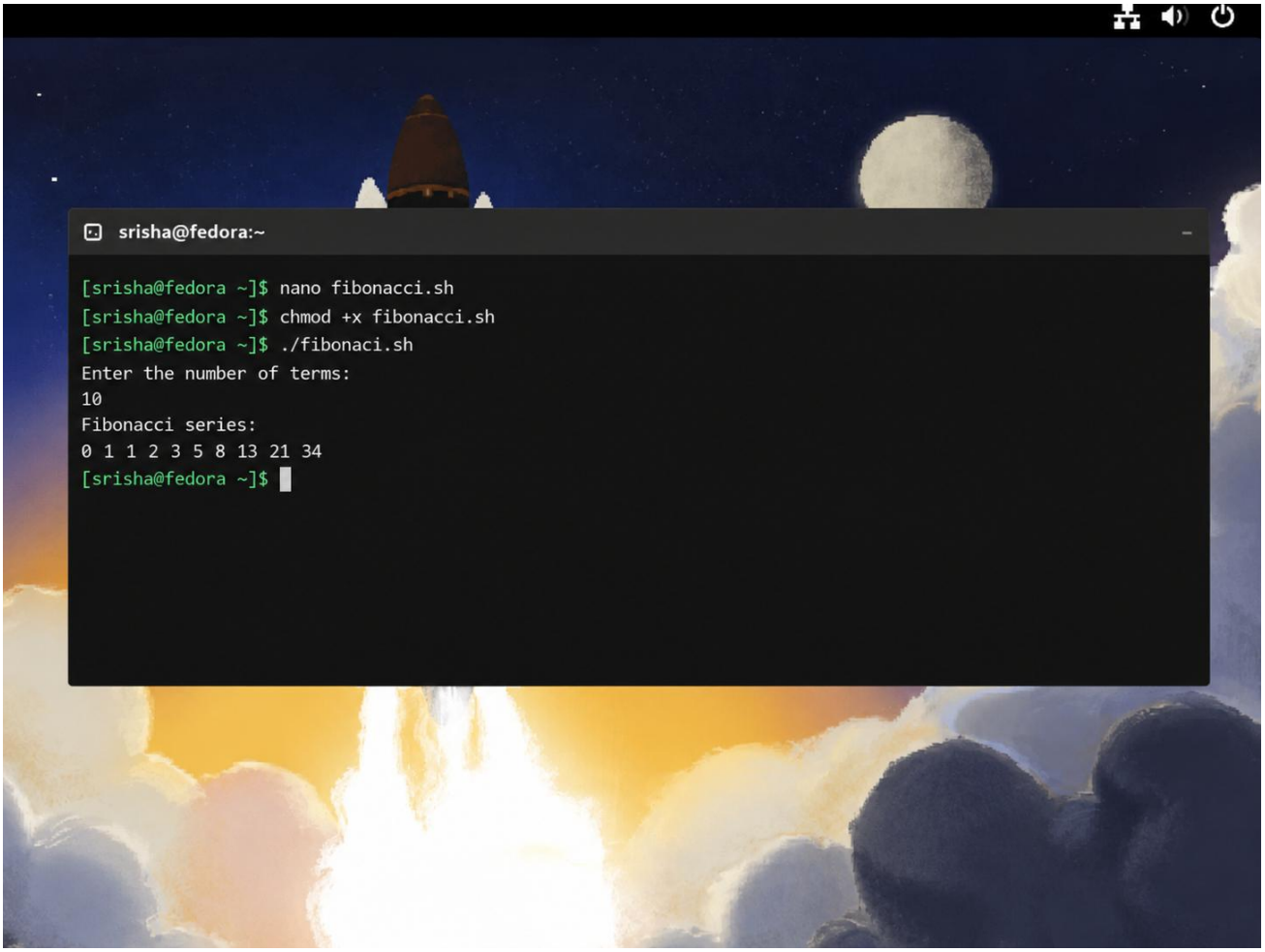
1. Create fibonacci.sh.
2. Enter the program and save the file.
3. Make the file executable and run it.
4. Enter the required number of terms.

PROGRAM:

```
echo "Enter number of terms:"
read n
a=0
b=1
for ((i=0; i<n; i++)); do
    echo -n "$a "
    next=$((a + b))
    a=$b
    b=$next
done
echo
```

OUTPUT:

```
Enter number of terms:
8
0 1 1 2 3 5 8 13
```



RESULT: Thus, the Fibonacci series was generated successfully using a for loop.